# SENTINEL-GNSS — Colab Training Notebook

**Before running:** Runtime ▸ Change runtime type ▸ **T4 GPU**

| What lives where | |
|---|---|
| Code + all data CSVs | GitHub (cloned in Step 1) |
| Feature windows (.npz) | Built fresh each session from CSV |
| Checkpoints + figures | Google Drive (survives disconnects) |

**If Colab disconnects mid-training:** Re-run Steps 1–3, then run the recovery cell at the bottom — it resumes from the last saved checkpoint automatically.

## Step 1 — Clone repo from GitHub

All code and processed CSVs (including the labelled dataset) are pulled directly from GitHub.

```
import os; os.chdir('/content')
```

```
import os, shutil

GITHUB_REPO = 'https://github.com/Jorshuare/AI-Based-Prediction-for-GNSS-Signal-Degradation.git'
REPO_DIR    = '/content/sentinel-gnss'

# Always anchor to /content first so the kernel is never left in a deleted directory
%cd /content

if os.path.exists(f'{REPO_DIR}/.git'):
    print('Repo already cloned — pulling latest ...')
    %cd {REPO_DIR}
    !git pull
else:
    # Remove stale directory if it exists without .git (e.g. leftover from failed clone)
    if os.path.exists(REPO_DIR):
        print('Removing stale directory ...')
        shutil.rmtree(REPO_DIR)
    print('Cloning repo ...')
    !git clone {GITHUB_REPO} {REPO_DIR}
    %cd {REPO_DIR}

print(f'\nWorking directory: {os.getcwd()}')
!git log --oneline -5

# Confirm labelled CSV came down with the repo
csv_path = f'{REPO_DIR}/data/labelled/sentinel_gnss_labelled.csv'
assert os.path.exists(csv_path), f'CSV not found at {csv_path}'
import pandas as pd
df = pd.read_csv(csv_path)
print(f'\nDataset: {len(df):,} rows × {len(df.columns)} columns — ready.')
```

```
/content
Cloning repo ...
Cloning into '/content/sentinel-gnss'...
remote: Enumerating objects: 465, done.
remote: Counting objects: 100% (142/142), done.
remote: Compressing objects: 100% (112/112), done.
```

```
remote: Total 465 (delta 58), reused 106 (delta 29), pack-reused 323 (from 1)
Receiving objects: 100% (465/465), 275.55 MiB | 23.46 MiB/s, done.
Resolving deltas: 100% (137/137), done.
Updating files: 100% (256/256), done.
/content/sentinel-gnss

Working directory: /content/sentinel-gnss
c4db656 (HEAD -> main, origin/main, origin/HEAD) fix: update model
4cddb00 fix: update model
c2d50f0 fix: balanced val subset for early stopping (500/class vs 97% CLEAN full val)
59c3212 fix: aux loss skipped for ablation models; baselines table handles flat metrics JSON
0aef1cd feat: aux t+0 head, warning weight [1,4,3], threshold tuning, drone cap, no-smote training

Dataset: 97,393 rows × 41 columns — ready.
```

## Step 2 — Install extra dependencies + verify GPU

Colab already ships with PyTorch, NumPy, pandas, scikit-learn, matplotlib, seaborn, scipy.
Only `imbalanced-learn` (SMOTE) needs installing.

```
!pip install -q imbalanced-learn

import torch
print(f'PyTorch  : {torch.__version__}')
print(f'CUDA     : {torch.cuda.is_available()}')
if torch.cuda.is_available():
    props = torch.cuda.get_device_properties(0)
    print(f'GPU      : {torch.cuda.get_device_name(0)}')
    print(f'VRAM     : {props.total_memory / 1e9:.1f} GB')
else:
    print('WARNING: No GPU — go to Runtime > Change runtime type > T4 GPU')
```

```
PyTorch  : 2.10.0+cu128
CUDA     : True
GPU      : Tesla T4
VRAM     : 15.6 GB
```

## Step 3 — Mount Google Drive (checkpoints + figures only)

Checkpoints are saved to Drive so training survives disconnects.
Data CSVs are **not** needed here — they came from GitHub in Step 1.

```
from google.colab import drive
import os, shutil

drive.mount('/content/drive')

REPO_DIR   = '/content/sentinel-gnss'
DRIVE_BASE = '/content/drive/MyDrive/sentinel-gnss'

# Folders that must persist across sessions
PAIRS = [
    (f'{REPO_DIR}/results/models/checkpoints', f'{DRIVE_BASE}/checkpoints'),
    (f'{REPO_DIR}/results/figures',            f'{DRIVE_BASE}/figures'),
]

for local, on_drive in PAIRS:
    os.makedirs(on_drive, exist_ok=True)
```

```
    os.makedirs(os.path.dirname(local), exist_ok=True)
    if os.path.isdir(local) and not os.path.islink(local):
        shutil.rmtree(local)   # remove empty local dir
    if not os.path.islink(local):
        os.symlink(on_drive, local)
    print(f'  {os.path.basename(local):15s} → {on_drive}')

# Report any existing checkpoints (from a previous run)
ckpt_dir = f'{DRIVE_BASE}/checkpoints'
ckpts = sorted(f for f in os.listdir(ckpt_dir) if f.endswith('.pt'))
if ckpts:
    print(f'\nExisting checkpoints on Drive ({len(ckpts)}):')
    for c in ckpts:
        print(f'  {c}')
    print('\nRun Step 5 with --resume to continue training.')
else:
    print('\nNo existing checkpoints — will start fresh.')
```

```
Mounted at /content/drive
  checkpoints     → /content/drive/MyDrive/sentinel-gnss/checkpoints
  figures         → /content/drive/MyDrive/sentinel-gnss/figures

No existing checkpoints — will start fresh.
```

## Step 4 — Build feature windows

Reads `data/labelled/sentinel_gnss_labelled.csv` → produces sliding-window tensors.
**Skips automatically if windows already exist** (safe to re-run).

```
%cd /content/sentinel-gnss
# Build SMOTE windows (used by baselines) — --force ensures y_0s label is included
!python -m src.models.feature_prep --force

# Build no-SMOTE windows (used by deep model — SMOTE creates incoherent temporal windows)
!python -m src.models.feature_prep --no_smote --force

# Sanity check
import numpy as np
print('\nWindow shapes (SMOTE — for baselines):')
for split in ('train', 'val', 'test'):
    d = np.load(f'data/processed/windows/{split}.npz')
    y0_label = "y_0s present" if "y_0s" in d else "y_0s MISSING"
    print(f'  {split:5s}  X={d["X"].shape}  CLEAN={np.sum(d["y_5s"]==0):,}  WARNING={np.sum(d["y_5s"]==1):,}  DEGRADED={np.sum(d["y_5s"]==2):,}  [{y0_label}]')

print('\nWindow shapes (no-SMOTE — for deep model):')
for split in ('train', 'val', 'test'):
    d = np.load(f'data/processed/windows_no_smote/{split}.npz')
    y0_label = "y_0s present" if "y_0s" in d else "y_0s MISSING"
    print(f'  {split:5s}  X={d["X"].shape}  CLEAN={np.sum(d["y_5s"]==0):,}  WARNING={np.sum(d["y_5s"]==1):,}  DEGRADED={np.sum(d["y_5s"]==2):,}  [{y0_label}]')
```

```
INFO    label dist: {0: 73370, 1: 15075, 2: 8948}
INFO    split dist: {'train': 53077, 'val': 26525, 'test': 17791}
INFO    Excluded 'supervisor_drone_1': 2,769 rows removed
INFO    Excluded 'supervisor_drone_2': 2,792 rows removed
INFO    Excluded 'supervisor_drone_12': 5,562 rows removed
INFO  Feature columns (34): ['alt', 'baseline_sats', 'clock_bias', 'cnr_trend', 'cnr_variance', 'cycle_slips', 'dop_ratio', 'elevation_violations', 'fix_continuity', 'fix_transitions', 'gdop', 'hdop',
INFO    Scaler saved → /content/sentinel-gnss/data/processed/scaler.pkl
```

```
INFO    test : 17,803 windows  | 5s label dist: [13740  3003  1060]
INFO  Applying SMOTE to training set …
INFO    SMOTE input  shape: (52757, 1020) | labels: [35223 10600  6934]
INFO    SMOTE output shape: (105669, 1020) | labels: [35223 35223 35223]
INFO    Saved train windows → /content/sentinel-gnss/data/processed/windows/train.npz  (105,669 samples)
INFO    Saved val   windows → /content/sentinel-gnss/data/processed/windows/val.npz  (12,896 samples)
INFO    Saved test  windows → /content/sentinel-gnss/data/processed/windows/test.npz  (17,803 samples)

=== Window summary ===
  train  X=(105669, 30, 34)  y_5s=[35223 35223 35223]
  val    X=(12896, 30, 34)  y_5s=[12089   444   363]
  test   X=(17803, 30, 34)  y_5s=[13740  3003  1060]
INFO  Excluding sources: ['supervisor_drone_1', 'supervisor_drone_2', 'supervisor_drone_12']
INFO  === --no_smote: saving class-weight-only windows to windows_no_smote/ ===
INFO  Loading /content/sentinel-gnss/data/labelled/sentinel_gnss_labelled.csv …
INFO    97,393 rows × 41 columns
INFO    label dist: {0: 73370, 1: 15075, 2: 8948}
INFO    split dist: {'train': 53077, 'val': 26525, 'test': 17791}
INFO    Excluded 'supervisor_drone_1': 2,769 rows removed
INFO    Excluded 'supervisor_drone_2': 2,792 rows removed
INFO    Excluded 'supervisor_drone_12': 5,562 rows removed
INFO  Feature columns (34): ['alt', 'baseline_sats', 'clock_bias', 'cnr_trend', 'cnr_variance', 'cycle_slips', 'dop_ratio', 'elevation_violations', 'fix_continuity', 'fix_transitions', 'gdop', 'hdop',
INFO    Scaler saved → /content/sentinel-gnss/data/processed/scaler.pkl
WARNING    Session 'scenario_a_r1' too short; skipping.
WARNING    Session 'scenario_a_r2' too short; skipping.
INFO    train: 52,757 windows  | 5s label dist: [35223 10600  6934]
INFO    val  : 12,896 windows  | 5s label dist: [12089   444   363]
INFO    test : 17,803 windows  | 5s label dist: [13740  3003  1060]
INFO    Saved train windows → /content/sentinel-gnss/data/processed/windows_no_smote/train.npz  (52,757 samples)
INFO    Saved val   windows → /content/sentinel-gnss/data/processed/windows_no_smote/val.npz  (12,896 samples)
INFO    Saved test  windows → /content/sentinel-gnss/data/processed/windows_no_smote/test.npz  (17,803 samples)

=== Window summary ===
  train  X=(52757, 30, 34)  y_5s=[35223 10600  6934]
  val    X=(12896, 30, 34)  y_5s=[12089   444   363]
  test   X=(17803, 30, 34)  y_5s=[13740  3003  1060]

Window shapes (SMOTE — for baselines):
  train  X=(105669, 30, 34)  CLEAN=35,223  WARNING=35,223  DEGRADED=35,223  [y_0s present]
  val    X=(12896, 30, 34)  CLEAN=12,089  WARNING=444  DEGRADED=363  [y_0s present]
  test   X=(17803, 30, 34)  CLEAN=13,740  WARNING=3,003  DEGRADED=1,060  [y_0s present]

Window shapes (no-SMOTE — for deep model):
  train  X=(52757, 30, 34)  CLEAN=35,223  WARNING=10,600  DEGRADED=6,934  [y_0s present]
  val    X=(12896, 30, 34)  CLEAN=12,089  WARNING=444  DEGRADED=363  [y_0s present]
  test   X=(17803, 30, 34)  CLEAN=13,740  WARNING=3,003  DEGRADED=1,060  [y_0s present]
```

```
!pip install -q xgboost
!python -m src.models.baselines
```

```
INFO   Loading windows …
INFO     Train: (105669, 1020)   Test: (17803, 1020)
INFO
── Tier 1: Majority-class ──────────────────────────────
INFO
── Tier 2: C/N₀ threshold rule ─────────────────────────
INFO
── Tier 3: Random Forest ───────────────────────────────
INFO     RandomForest  +5s  MacroF1=0.5443  MCC=0.4171
INFO     RandomForest  +15s  MacroF1=0.5327  MCC=0.4025
INFO     RandomForest  +30s  MacroF1=0.5183  MCC=0.3813
INFO
── Tier 3: XGBoost ─────────────────────────────────────
INFO     XGBoost       +5s  MacroF1=0.5543  MCC=0.4296
INFO     XGBoost       +15s  MacroF1=0.5335  MCC=0.4055
INFO     XGBoost       +30s  MacroF1=0.5206  MCC=0.3902
INFO
```

```
INFO   ================================================================================
INFO     Comparison Table — Test Set Macro-F1  (higher is better)
INFO   ================================================================================
INFO     Method                      Justification                +5s     +15s     +30s
INFO   --------------------------------------------------------------------------------
INFO     MajorityClass               Trivial floor              0.2904   0.0377   0.0380
INFO     CNR_Threshold               Domain rule (RTCM)         0.0375   0.0377   0.0380
INFO     RandomForest                Classical ML baseline      0.5443   0.5327   0.5183
INFO     XGBoost                     Classical ML baseline      0.5543   0.5335   0.5206
INFO   ================================================================================
INFO     Macro-F1: equally weights CLEAN / WARNING / DEGRADED classes.
INFO     Random 3-class predictor would score ≈ 0.3333.
INFO   ================================================================================
INFO
Full metrics saved → /content/sentinel-gnss/results/baselines/baseline_comparison.json
```

## Step 5 — Train

`--resume` continues from the last Drive checkpoint if it exists, otherwise starts fresh.
Checkpoints are saved directly to Drive every 10 epochs + whenever a new best val F1 is reached.

```
import shutil, os, glob

# Wipe ALL checkpoints from Drive so previous runs don't interfere
drive_ckpt = '/content/drive/MyDrive/sentinel-gnss/checkpoints'
local_ckpt = '/content/sentinel-gnss/results/models/checkpoints'

for ckpt_path in [drive_ckpt, local_ckpt]:
    if os.path.exists(ckpt_path):
        for f in glob.glob(f'{ckpt_path}/*'):
            os.remove(f)
        print(f'Cleared: {ckpt_path}')

%cd /content/sentinel-gnss
# Deep model trained WITHOUT SMOTE.
# Class weights [1.0, 4.0, 3.0] handle imbalance (WARNING boosted most — lowest precision).
# Auxiliary t+0 head forces model to learn current state as prerequisite for prediction.
!python -m src.models.train --batch_size 256 --window_dir data/processed/windows_no_smote
```

```
Cleared: /content/drive/MyDrive/sentinel-gnss/checkpoints
Cleared: /content/sentinel-gnss/results/models/checkpoints
/content/sentinel-gnss
INFO  Device: cuda
INFO    GPU: Tesla T4
INFO  Loading windows …
INFO    train batches: 207  val batches: 51  val_stop (balanced): 1089 samples
/content/sentinel-gnss/src/models/transformer_lstm.py:211: UserWarning: enable_nested_tensor is True, but self.use_nested_tensor is False because encoder_layer.norm_first was True
  self.transformer = nn.TransformerEncoder(
SentinelGNSS  |  parameters: 367,692
INFO
============================================================
INFO    Training SENTINEL-GNSS  |  367,692 parameters
INFO    Epochs: 0 → 150  |  Patience: 30
INFO    class_weights=[1.0, 2.0, 1.5]  focal_gamma=1.0  label_smoothing=0.1  dropout=0.2
INFO  ============================================================

INFO  Epoch 001/150  | tr_loss=0.4519  val_loss=0.1534  | val_F1(5s/15s/30s): 0.813/0.791/0.760  | stop_F1=0.793  | lr=4.00e-04  6.7s
INFO    ★ New best stop macro-F1 = 0.7926  → checkpoint_best.pt
INFO  Epoch 002/150  | tr_loss=0.2866  val_loss=0.1424  | val_F1(5s/15s/30s): 0.801/0.781/0.748  | stop_F1=0.804  | lr=6.00e-04  5.5s
INFO    ★ New best stop macro-F1 = 0.8035  → checkpoint_best.pt
INFO  Epoch 003/150  | tr_loss=0.2582  val_loss=0.1480  | val_F1(5s/15s/30s): 0.816/0.770/0.743  | stop_F1=0.815  | lr=8.00e-04  6.2s
INFO    ★ New best stop macro-F1 = 0.8145  → checkpoint_best.pt
```

```
INFO  Epoch 004/150  | tr_loss=0.2390  val_loss=0.1388  | val_F1(5s/15s/30s): 0.817/0.792/0.783  | stop_F1=0.827  | lr=1.00e-03  4.9s
INFO    ★ New best stop macro-F1 = 0.8273  → checkpoint_best.pt
INFO  Epoch 005/150  | tr_loss=0.2270  val_loss=0.1244  | val_F1(5s/15s/30s): 0.848/0.824/0.784  | stop_F1=0.846  | lr=1.00e-03  6.1s
INFO    ★ New best stop macro-F1 = 0.8457  → checkpoint_best.pt
INFO  Epoch 006/150  | tr_loss=0.2144  val_loss=0.1375  | val_F1(5s/15s/30s): 0.822/0.813/0.718  | stop_F1=0.813  | lr=1.00e-03  5.1s
INFO  Epoch 007/150  | tr_loss=0.2077  val_loss=0.1407  | val_F1(5s/15s/30s): 0.825/0.780/0.649  | stop_F1=0.781  | lr=1.00e-03  5.0s
INFO  Epoch 008/150  | tr_loss=0.2027  val_loss=0.1344  | val_F1(5s/15s/30s): 0.843/0.809/0.794  | stop_F1=0.839  | lr=9.99e-04  6.6s
INFO  Epoch 009/150  | tr_loss=0.1950  val_loss=0.1379  | val_F1(5s/15s/30s): 0.828/0.803/0.720  | stop_F1=0.816  | lr=9.98e-04  5.1s
INFO  Epoch 010/150  | tr_loss=0.1906  val_loss=0.1533  | val_F1(5s/15s/30s): 0.823/0.767/0.659  | stop_F1=0.796  | lr=9.97e-04  6.3s
INFO    → Periodic checkpoint: checkpoint_epoch_010.pt
INFO  Epoch 011/150  | tr_loss=0.1838  val_loss=0.1514  | val_F1(5s/15s/30s): 0.818/0.795/0.730  | stop_F1=0.818  | lr=9.96e-04  5.0s
INFO  Epoch 012/150  | tr_loss=0.1779  val_loss=0.1504  | val_F1(5s/15s/30s): 0.809/0.751/0.701  | stop_F1=0.797  | lr=9.94e-04  5.2s
INFO  Epoch 013/150  | tr_loss=0.1734  val_loss=0.1491  | val_F1(5s/15s/30s): 0.796/0.779/0.767  | stop_F1=0.830  | lr=9.93e-04  6.3s
INFO  Epoch 014/150  | tr_loss=0.1691  val_loss=0.1499  | val_F1(5s/15s/30s): 0.816/0.783/0.763  | stop_F1=0.831  | lr=9.91e-04  5.1s
INFO  Epoch 015/150  | tr_loss=0.1649  val_loss=0.1556  | val_F1(5s/15s/30s): 0.783/0.774/0.767  | stop_F1=0.829  | lr=9.88e-04  6.5s
INFO  Epoch 016/150  | tr_loss=0.1602  val_loss=0.1454  | val_F1(5s/15s/30s): 0.808/0.795/0.777  | stop_F1=0.854  | lr=9.86e-04  4.9s
INFO    ★ New best stop macro-F1 = 0.8541  → checkpoint_best.pt
INFO  Epoch 017/150  | tr_loss=0.1570  val_loss=0.1584  | val_F1(5s/15s/30s): 0.834/0.785/0.754  | stop_F1=0.827  | lr=9.83e-04  5.1s
INFO  Epoch 018/150  | tr_loss=0.1555  val_loss=0.1637  | val_F1(5s/15s/30s): 0.803/0.784/0.750  | stop_F1=0.826  | lr=9.80e-04  6.1s
INFO  Epoch 019/150  | tr_loss=0.1525  val_loss=0.1734  | val_F1(5s/15s/30s): 0.811/0.738/0.650  | stop_F1=0.776  | lr=9.77e-04  4.8s
INFO  Epoch 020/150  | tr_loss=0.1487  val_loss=0.1449  | val_F1(5s/15s/30s): 0.830/0.798/0.777  | stop_F1=0.837  | lr=9.74e-04  6.2s
INFO    → Periodic checkpoint: checkpoint_epoch_020.pt
INFO  Epoch 021/150  | tr_loss=0.1476  val_loss=0.1614  | val_F1(5s/15s/30s): 0.819/0.770/0.745  | stop_F1=0.845  | lr=9.70e-04  5.2s
INFO  Epoch 022/150  | tr_loss=0.1466  val_loss=0.1892  | val_F1(5s/15s/30s): 0.786/0.743/0.698  | stop_F1=0.821  | lr=9.66e-04  4.9s
INFO  Epoch 023/150  | tr_loss=0.1410  val_loss=0.1694  | val_F1(5s/15s/30s): 0.814/0.755/0.715  | stop_F1=0.810  | lr=9.62e-04  6.3s
INFO  Epoch 024/150  | tr_loss=0.1382  val_loss=0.1887  | val_F1(5s/15s/30s): 0.801/0.737/0.613  | stop_F1=0.767  | lr=9.58e-04  4.8s
INFO  Epoch 025/150  | tr_loss=0.1367  val_loss=0.1695  | val_F1(5s/15s/30s): 0.792/0.769/0.742  | stop_F1=0.812  | lr=9.54e-04  6.0s
INFO  Epoch 026/150  | tr_loss=0.1348  val_loss=0.1901  | val_F1(5s/15s/30s): 0.799/0.759/0.668  | stop_F1=0.799  | lr=9.49e-04  5.3s
INFO  Epoch 027/150  | tr_loss=0.1327  val_loss=0.1827  | val_F1(5s/15s/30s): 0.796/0.729/0.665  | stop_F1=0.788  | lr=9.44e-04  4.9s
INFO  Epoch 028/150  | tr_loss=0.1294  val_loss=0.1715  | val_F1(5s/15s/30s): 0.814/0.776/0.721  | stop_F1=0.802  | lr=9.39e-04  6.2s
INFO  Epoch 029/150  | tr_loss=0.1286  val_loss=0.1850  | val_F1(5s/15s/30s): 0.795/0.739/0.673  | stop_F1=0.793  | lr=9.34e-04  4.9s
INFO  Epoch 030/150  | tr_loss=0.1268  val_loss=0.1751  | val_F1(5s/15s/30s): 0.812/0.755/0.720  | stop_F1=0.812  | lr=9.28e-04  5.8s
INFO    → Periodic checkpoint: checkpoint_epoch_030.pt
INFO  Epoch 031/150  | tr_loss=0.1254  val_loss=0.1723  | val_F1(5s/15s/30s): 0.806/0.759/0.714  | stop_F1=0.805  | lr=9.23e-04  5.5s
INFO  Epoch 032/150  | tr_loss=0.1222  val_loss=0.1909  | val_F1(5s/15s/30s): 0.766/0.743/0.701  | stop_F1=0.810  | lr=9.17e-04  5.0s
```

## Step 6 — Evaluate

Loads `checkpoint_best.pt` from Drive, runs all 14 analyses, saves figures to Drive.

```
%cd /content/sentinel-gnss
# --tune_thresholds: sweeps WARNING/DEGRADED probability thresholds on val set
#   to maximise macro-F1. Free improvement — no retraining needed.
# --window_dir: val/test splits are identical to SMOTE dir, but keeps paths consistent.
!python -m src.models.evaluate --tune_thresholds --window_dir data/processed/windows_no_smote
```

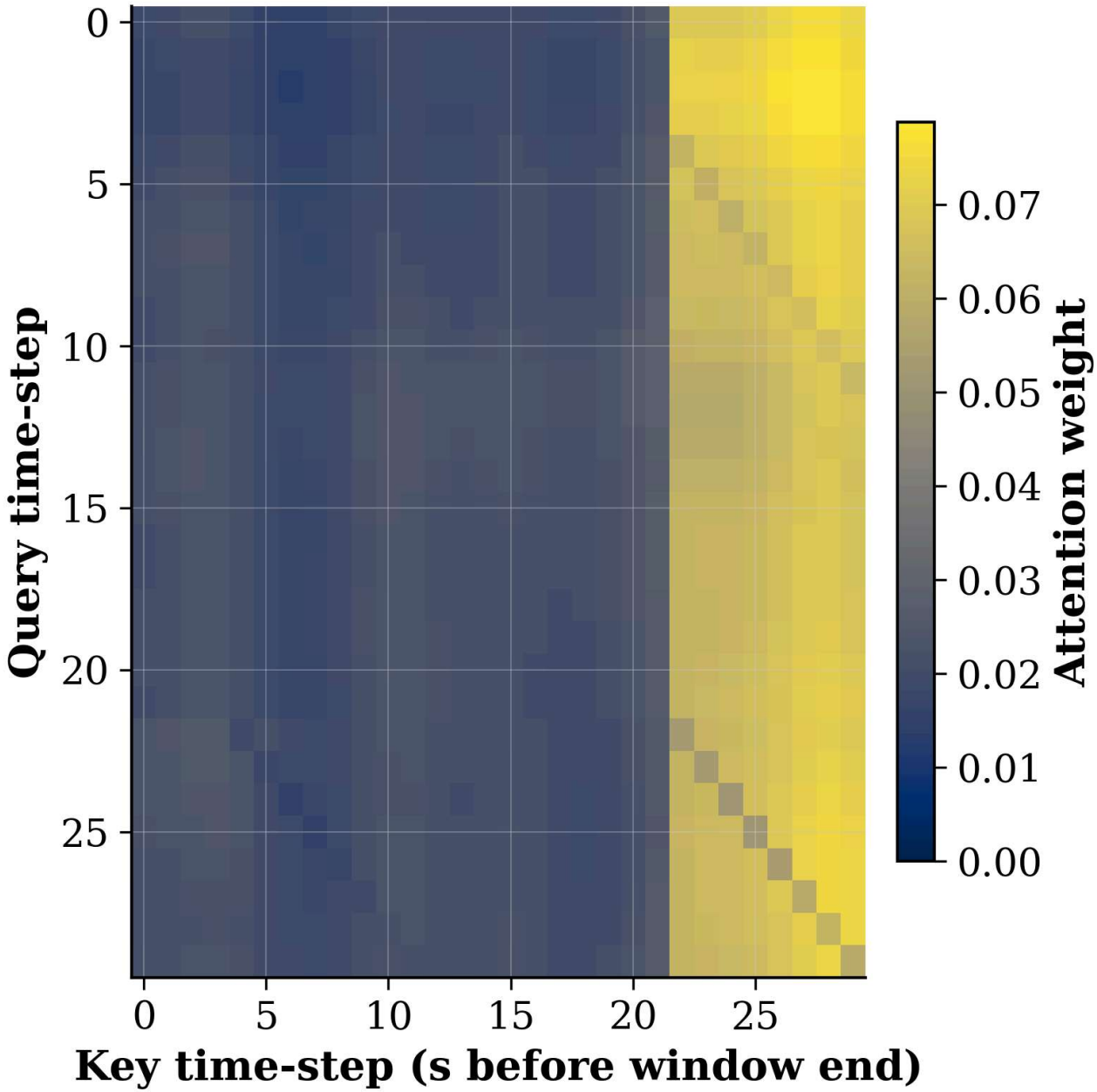
```
/content/sentinel-gnss
INFO  Device: cuda
/content/sentinel-gnss/src/models/transformer_lstm.py:211: UserWarning: enable_nested_tensor is True, but self.use_nested_tensor is False because encoder_layer.norm_first was True
  self.transformer = nn.TransformerEncoder(
SentinelGNSS  |  parameters: 367,692
INFO    Loaded model from checkpoint_best.pt  (epoch=16)
INFO    Running inference on 'test' split …
INFO    Tuning classification thresholds on val set …
INFO    Threshold +5s:  WARN=0.39  DEG=0.90  → val macro-F1=0.8270
INFO    Threshold +15s:  WARN=0.48  DEG=0.90  → val macro-F1=0.8075
INFO    Threshold +30s:  WARN=0.90  DEG=0.90  → val macro-F1=0.7885
INFO    Thresholds saved → /content/sentinel-gnss/results/figures/tuned_thresholds.json

  ── With tuned thresholds ──

──────────────────────────────────────────────────────────────────
  SENTINEL-GNSS Evaluation Results  |  split = test+tuned
──────────────────────────────────────────────────────────────────
Horizon      Acc    MacroF1     WtF1        κ      MCC
```

```
  +5s     0.5011  0.4306    0.5613  0.0856  0.1015
  +15s    0.4957  0.4076    0.5552  0.0695  0.0823
  +30s    0.4563  0.3223    0.5287  0.0754  0.0967
──────────────────────────────────────────────────────────────

  +5s per-class breakdown:
    CLEAN       P=0.836  R=0.529  F1=0.648  n=13740
    WARNING     P=0.144  R=0.370  F1=0.208  n=3003
    DEGRADED    P=0.381  R=0.509  F1=0.436  n=1060

  +15s per-class breakdown:
    CLEAN       P=0.828  R=0.529  F1=0.645  n=13710
    WARNING     P=0.148  R=0.384  F1=0.214  n=3025
    DEGRADED    P=0.346  R=0.383  F1=0.364  n=1068

  +30s per-class breakdown:
    CLEAN       P=0.817  R=0.519  F1=0.635  n=13667
    WARNING     P=0.355  R=0.132  F1=0.193  n=3061
    DEGRADED    P=0.079  R=0.588  F1=0.139  n=1075

──────────────────────────────────────────────────────────────

INFO    Using tuned-threshold predictions for figures.

──────────────────────────────────────────────────────────────
  SENTINEL-GNSS Evaluation Results  |  split = test
──────────────────────────────────────────────────────────────
Horizon      Acc    MacroF1     WtF1       κ      MCC
──────────────────────────────────────────────────────────────
  +5s     0.5011  0.4306    0.5613  0.0856  0.1015
  +15s    0.4957  0.4076    0.5552  0.0695  0.0823
  +30s    0.4563  0.3223    0.5287  0.0754  0.0967
──────────────────────────────────────────────────────────────

  +5s per-class breakdown:
    CLEAN       P=0.836  R=0.529  F1=0.648  n=13740
    WARNING     P=0.144  R=0.370  F1=0.208  n=3003
    DEGRADED    P=0.381  R=0.509  F1=0.436  n=1060
```

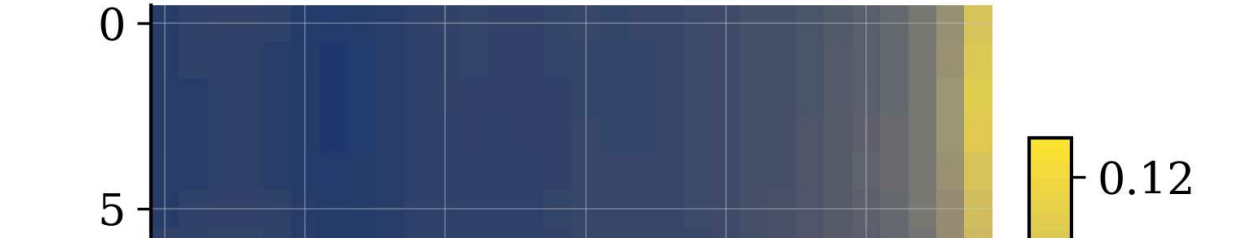
## Step 7 — View all figures inline

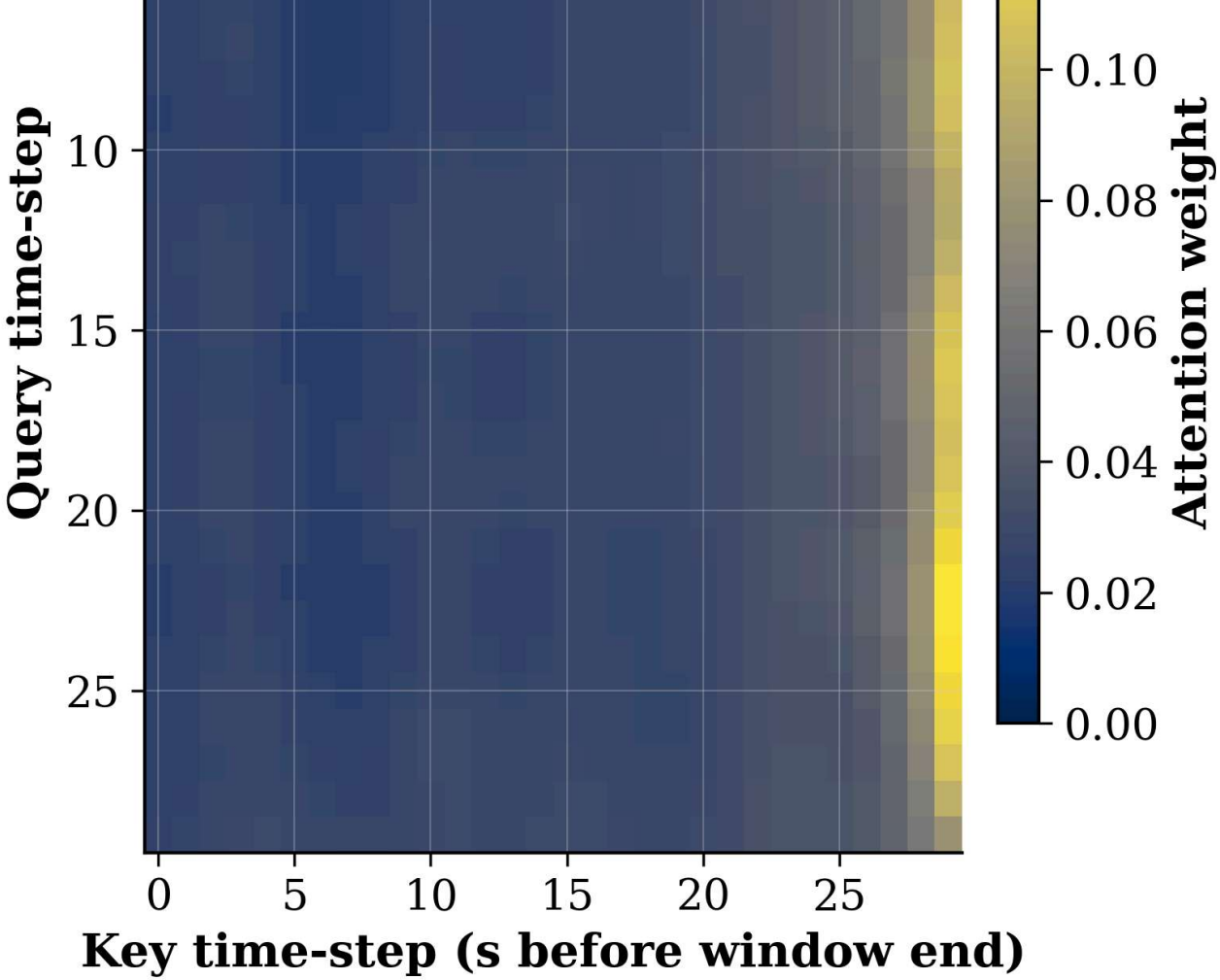
```
import glob
from IPython.display import Image, display

figs = sorted(glob.glob('/content/sentinel-gnss/results/figures/*.png'))
print(f'{len(figs)} figures saved to Drive:')
for f in figs:
    name = f.split('/')[-1]
    print(f'  {name}')
    display(Image(filename=f, width=950))
```

```
13 figures saved to Drive:
  attention_heatmap_clean.png
```




attention_heatmap_degraded.png






attention_heatmap_warning.png